

From Model to Prototype: A User-Centred Workflow for Cloud-Based AI-Assisted Chest X-Ray Analysis

1st Anna-Livia Martin

*Computer Science – Software and Systems Engineering
RheinMain University of Applied Sciences
Wiesbaden, Germany
anna-livia.martin@student.hs-rm.de*

2nd David Rossel

*Computer Science – Software and Systems Engineering
RheinMain University of Applied Sciences
Wiesbaden, Germany
david.rossel@student.hs-rm.de*

Abstract—In recent years, artificial intelligence (AI) has demonstrated considerable potential in medical image analysis, particularly in the detection of diseases on X-ray images. Despite these advances in model performance, the clinical use of such systems remains limited. This paper presents a prototype of a cloud-based system for the AI-assisted analysis of chest X-rays, integrated into a patient management system, which combines technical, user-centred and regulatory aspects. User-friendliness is investigated through a user study and assessed in relation to European regulatory requirements for software as a medical device (SaMD). The results demonstrate the feasibility of integrating these aspects into a unified prototype and underscore the importance of considering not only technical performance but also user-friendliness, transparency and integration into the workflow and compliance with regulatory requirements in scientific studies.

Index Terms—software as a medical device, azure, usability, machine learning, convolutional neural network, UX, UI, prototype

I. INTRODUCTION

Since the mid-2010s, convolutional neural networks (CNNs) have substantially advanced performance in medical image analysis, particularly in chest radiography classification and detection tasks [1], [2], and their diagnostic performance has continued to improve ever since [3]–[5].

Despite strong technical performance, the clinical adoption of AI-based diagnostic systems remains limited. A 2022 survey by the European Society of Radiology reported that only 40% of all radiologists questioned reported having any practical clinical experience of using AI systems, although a majority (75.7%) consider their outputs generally reliable [6]. Thus, although powerful models exist and radiologists largely consider AI to be reliable, actual clinical use is significantly lower than would be expected based on the reported model performance. The results of the surveys by Becker et al. [6] suggest that the capabilities of AI models are no longer the sole determining factor. Rather, most radiologists do not report a significant improvement in their clinical workflow, with some users reporting increased workload when using AI system [6].

A commonly cited barrier is the limited interpretability of machine learning (ML) models in clinical settings. Rafferty

et al. [7] argue that the ‘black-box’ nature of AI models in clinical use undermines the interpretability and trustworthiness of the models. In their study they examine the contribution of Explainable Artificial Intelligence (XAI) to the clinical interpretability of AI models and argue that clinically needed relevant explanations are often not provided by existing explanatory approaches of machine learning models. Further challenges relate to ethical issues [8]. Raymond Geis et al. [8] found that current research often fails to adequately address ethical issues such as upholding patient-centred ethical principles. Explainability is therefore no longer a problem to be solved exclusively through technical means. Rather, it also encompasses equally important ethical, social and legal aspects [9].

Coiera [10] identifies further causes for the low level of clinical acceptance, including the fact that user interfaces are inadequately adapted to clinical workflows and, as a result, helpful systems are not used. Similarly, AI-systems are often not integrated into the workflow making their usage less likely [10]. Added to this are the stringent regulatory requirements imposed by various institutions, which can vary from country to country at an international level [11]–[14] and can therefore only be replicated to a limited extent within scientific research projects. This results in a significant discrepancy between regulatory requirements and typical scientific research projects.

The reason for the underutilisation of AI models therefore often lies not solely in the quality of the models, but in a lack of usability, transparency, regulatory requirements and integration into clinical workflows. Whilst many papers highlight individual aspects of these challenges and attempt to improve them, such as the accuracy of ML models [3], [15], [16], XAI [17]–[19], regulatory compliance [12], [20], [21] or user acceptance [10], [22]–[24], comparatively few studies have systematically integrated these dimensions into a unified perspective [22], [25]. This fragmentation contributes to a persistent gap between high-performing research prototypes and clinically deployable systems.

This paper addresses this gap by designing a prototype cloud-based system for AI-assisted analysis of chest X-rays.

We demonstrate how existing technical methods can be integrated into a user-centred workflow. The feasibility and perceived usability of the approach are investigated through a within-subjects user study.

Section II provides an overview of the current state of the art in the various aspects of software as a medical device. Chapter III begins with a description of the concept behind the developed prototype and outlines the underlying design goals as well as the user-centered workflow. Subsequently, Chapter IV describes the system architecture. This is followed by Chapter V, which examines the design of the user interface in greater detail. Building on this, Chapter VI describes the technical components and their implementation. The cloud-based deployment of the prototype is explained in Chapter VI-D. Chapter VIII then examines the relevant regulatory requirements within the European Union, specifically for the prototype and generally for software as a medical device. This is followed by a discussion of the findings in Chapter IX. Finally, Chapter X lists possible limitations and opportunities for further research, followed by a conclusion.

II. BACKGROUND AND RELATED WORK

This chapter provides an overview of the current state of research on AI-based systems in medical image analysis and examines the technical, explainable, and user-centered requirements, as well as global regulatory aspects, that are relevant to their clinical application.

A. Machine Learning for Chest X-ray Analysis

The automatic analysis of (medical) image data is one of the largest real-world applications in the field of machine learning. In particular, deep neural networks have led to significant improvements in performance in the past when it comes to detecting pathological findings in chest X-rays. A major milestone in the field of disease detection from X-ray images was the work by Rajpurkar et al. [1]. With CheXNet, they introduced a convolutional neural network model that was able to achieve a diagnostic performance for pneumonia on par (and above) with that of radiologists. The review by Litjens et al. [2] summarizes these early developments and shows that deep learning already demonstrated significant potential for medical disease detection about 10 years ago.

The performance of such early models was continuously improved in the years that followed [3]–[5], [15]. Recent work in the field of disease detection in chest X-rays demonstrates high diagnostic accuracy for various clinical conditions [3]–[5], [15]. At the same time, early studies demonstrate that the multi-class classification of clinical conditions is more complex than the detection of a single disease [16].

A key driver of this development is, in particular, the availability of large datasets on which these models can be trained. Datasets such as NIH ChestX-ray [26], CheXpert [27], MIMIC-CXR [28], PadChest [29], and the pneumonia dataset provided by Shih et al. [30] form the basis for numerous research projects. At the same time, the diversity of these datasets also poses a challenge. For example, they feature

different clinical presentations, labeling systems, sizes, and annotations. Conversely, this leads to research studies addressing different research questions and models being developed with varying diagnostic capabilities. This makes direct comparisons difficult. Nakagawa et al. [5] also point out that current studies often examine different disease spectra and thus pursue different clinical objectives.

These recent research developments show that diagnostic performance continues to improve. However, current research efforts remain primarily focused on developing new, more powerful models. Topics such as integration into clinical workflows, regulatory requirements, or the usability of the developed systems, on the other hand, are often evaluated to a lesser extent.

B. Explainable AI in Medical Imaging

As machine learning models become increasingly capable of identifying diseases, it also becomes more vital to present those decision-making processes in a way that is transparent. In this context, neural networks are considered “black boxes” that offer no insight into their own decision-making processes, making it impossible to determine which input leads to which output and why. In the field of diagnostic medicine, this in particular poses a special challenge [7].

The research by Barredo et al. [31] is considered one of the most fundamental works in the field of Explainable AI (XAI). Barredo et al. [31] distinguish between several closely related terms. Transparency and interpretability refer here to a model’s passive ability to be understandable to a human. Explainability, on the other hand, refers to the active counterpart to this. This section focuses on the specific actions taken to make the model understandable to the user. The goal of explainable artificial intelligence is to present the model’s internal decision-making processes in a way that is understandable to the respective target audience, thereby promoting transparency and trust [31].

In the field of XAI, various approaches for explaining individual models have been proposed in recent years [17]–[19]. Examples include the XAI approaches by Ribeiro et al. [17] and Selvaraju et al. [18]. Ribeiro et al. introduce LIME, an explanation system that provides model-independent local explanations for any classification model. Selvaraju et al. [18] present Grad-CAM, a method specifically designed for convolutional neural networks that can identify which input regions are critical for the model’s predictions. Tjoa et al. [19] provide a more comprehensive overview of various approaches.

Despite this progress, the practical value of many explanation methods remains controversial. For example, Rafferty et al. [7] argue that existing explanations for predictions based on chest X-rays are insufficient, as current explanations often fail to provide the clinically relevant insights necessary for practical application. Their findings highlight the importance of involving domain experts in the development of XAI in the medical field.

Overall, research shows that many different approaches to mitigating black boxes already exist. However, it remains

unclear to what extent these approaches meet the requirements of everyday clinical practice and to what extent they can be integrated into medical software.

C. Clinical Decision Support Systems and User Acceptance

Clinical decision support systems assist healthcare professionals in interpreting, diagnosing, and treating patients [22]. The goal of these systems is to relieve the burden on treating physicians, who are the sole processors of information, by providing information relevant to medical decisions. Consequently, they are intended to lead to a reduction in the error rate [22]. Kheirat et al. distinguish between two different types of clinical decision support systems (CDSS): knowledge-based and data-driven. While knowledge-based systems primarily provide treatment-related information and rely on expert knowledge, data-driven systems use machine learning to identify patterns in historical patient data [22].

Khairat et al. [22] suggest that the low adoption rate of data-driven systems may be due, among other things, to a lack of user acceptance. According to them, high diagnostic value alone is not sufficient if the expectations and requirements of future users are not taken into account. This ties in with the Technology Acceptance Model (TAM), which was first described by Fred Davis [32] in 1993. The TAM identifies the perceived usefulness of a system and its perceived ease of use as key factors that influence attitudes toward the system and its actual use [32]. Building on this, Venkatesh et al. [33] developed the Unified Theory of Acceptance and Use of Technology (UTAUT) model. In their research, they demonstrate that performance expectancy is the strongest factor influencing the acceptance of a technology.

There are also various factors that influence user acceptance and system usage in medical systems [10], [23]. Coiera [10] cites, among other things, insufficient integration into existing work processes and a lack of user-friendliness in the user interface as reasons why CDSSs are often not used at all or only to a limited extent, despite their capabilities. Wendt et al. [23] describe additional factors, such as the relevance of the information presented, the perceived validity of the data, and the effort and time required, as factors influencing the use of CDSS. Johnson et al. [34] confirm these relationships in a long-term study and show that, in particular, perceived usefulness and user-friendliness have a positive influence on the long-term use of CDSS. In contrast, a lack of computer literacy and a lack of optimism can have a negative impact on system usage and satisfaction [34]. Furthermore, Van Schaik et al. [24] argue that physicians often evaluate new systems more on the basis of their expected benefits than on their previous experiences with computer systems.

In addition to functional aspects, the design of the user interface also plays a crucial role [35]. Norman [35] argues that interactive systems should be self-explanatory so that users do not feel overwhelmed and errors in usage can be avoided. He also argues that intuitive design leads to more frequent use. We argue that such considerations can be directly applied to AI-powered CDSSs, whose acceptance depends not only on

diagnostic performance but also on a user interface that is easy to understand and integrated into existing workflows.

Overall, existing research shows that the successful implementation of AI-based CDSS goes far beyond the development of powerful machine-learning models. Rather, key factors include high user-friendliness, integration into clinical workflows, and consideration of the expectations of medical users.

D. Ethical and Regulatory Requirements

With the increasing use of artificial intelligence in healthcare, ethical and regulatory requirements are coming to the forefront alongside technical requirements. While many machine-learning models demonstrate significant improvements in diagnostic performance, their safe and responsible integration into clinical workflows remains a major hurdle.

Raymond Geis et al. [8] emphatically note that one goal of further developments should be the establishment of ethical requirements, specifically, explicit guidelines on how AI should be used in practice. They emphasize that it is essential to ensure that AI serves the patient's best interests and should not be driven by purely financial interests [8]. Amann et al. [9] further argue that explainability is not merely a technical challenge. Rather, it encompasses ethical, social, and legal issues that must more precisely define and regulate the interaction between humans and AI.

In addition to ethical considerations, there are a wide range of regulatory requirements for software intended for medical use. At the international level, such information systems are formalized and defined by the International Medical Device Regulators Forum (IMDRF) [21]. The IMDRF's goal is to harmonize regulatory requirements internationally. To this end, it publishes guiding principles for (AI- and ML-based or software-based) medical devices [21]. Among other things, these principles define the multidisciplinary expertise required throughout the entire product development process, the consideration of human-AI interaction, and the validation of existing systems under clinically relevant conditions [21].

Despite these efforts, there are still, in some cases, significant differences in the regulatory frameworks. While, for example, the U.S. FDA bases its definitions on the IMDRF's SaMD concept [36], in the European Union, the Medical Device Regulation (MDR) serves as the central legal framework for medical devices [37]. While, for example, in the European Union, "software as a medical device" is defined through the term "medical device" [37], there is no standalone definition for it in the United States. Instead, the term "medical device software" is used there to describe SaMD, and reference is made only to the IMDRF for its definitions of "software as a medical device" [36]. Supplementary guidelines from the European Commission further specify which software is to be classified as software as a medical device [38]. Together, these regulatory frameworks require extensive clinical evaluations, technical documentation, and quality assurance measures throughout the entire product lifecycle [37], but at the same time do not provide a uniform definition of terms.

The specific regulations of the European Union are examined in more detail in Chapter VIII and discussed in relation to the prototype. This poses particular challenges for scientific research projects. Odaibo [12] shows that even the terminology surrounding AI-based medical devices is complex and highly normative. As a result, the full implementation of regulatory requirements is often only possible to a limited extent within the context of academic research projects.

Overall, research shows that high-performance machine learning models alone are not sufficient to successfully integrate AI systems into clinical practice. In addition to diagnostic accuracy, ethical principles, regulatory requirements, and the safe and transparent use of these systems must also be taken into account.

To date, however, these research areas have been studied largely in isolation from one another, with ethics and regulations largely excluded from consideration. Research in the field of machine learning often focuses on optimizing model performance using established datasets. Research on Explainable Artificial Intelligence develops new explanation methods but generally considers their integration into clinical software systems only to a limited extent. Studies on user acceptance, on the other hand, primarily analyse organizational or user-related factors without actually implementing modern AI systems. Regulatory work defines requirements for safe medical devices but primarily describes normative frameworks rather than specific software architectures. Overall, there have been only a few studies to date that examine the development of AI-based software as a medical device from technical, user-centered, and regulatory perspectives collectively. In particular, there is a lack of prototype systems that combine powerful machine-learning models with a user-oriented interface, cloud-based deployment, and regulatory requirements, and that evaluate their practical usability. This raises the question of what a prototype integration of machine learning for chest X-rays with a cloud-based software as a medical device might look like, focusing on usability and regulatory requirements.

The main contributions of this work are threefold:

- 1) We propose and implement a user-centered workflow for cloud-based AI-assisted chest X-ray analysis that integrates existing machine learning and explainability techniques into a functional proof-of-concept prototype.
- 2) We provide an empirical evaluation of the prototype through a within-subjects user study, investigating perceived usability and usefulness using the Questionnaire for User Experience (QUE).
- 3) We translate relevant European regulatory requirements for software used as a medical device into concrete design considerations and demonstrate their application within the prototype.

III. CONCEPT

The goal of the prototype to be developed is to integrate, on the one hand, AI-supported analysis of chest X-rays and, on the other hand, a user-centered application for managing

patient-related information. To this end, the prototype combines existing methods for classifying diseases in chest X-rays with a visual explanation of the prediction using Grad-CAM and a web-based application for patient management. The goal is to come as close as possible to software used in real-world medical contexts while enabling seamless integration of AI.

A. Design Goals

The goal of the application's design is to integrate AI with patient management, with a particular focus on maximizing compliance with regulatory requirements as well as usability and cloud deployability. We therefore formulate these four design goals.

- Integration of existing AI methods: The application should integrate an existing ML model for disease detection in chest X-rays.
- Explainability: The AI results generated for disease detection should be presented in a way that is understandable to medical staff. To this end, the application will provide both a visual explanation of the detection and a confidence score associated with the classification.
- Input modalities for AI methods: It should be possible to initiate medical AI detection using image formats such as PNG and JPEG, as well as the medical X-ray format DICOM.
- User-centered interaction: The analysis should be capable of being performed within a patient-centered workflow.
- Regulatory-informed design: Requirements for software as a medical device are taken into account in the design and implementation of the application.

B. User-centred Workflow

Based on the defined design goals, a workflow was designed that combines patient management and AI analysis into a single workflow. The goal is explicitly not to describe the AI analysis as a separate step, but rather as an integrated aspect of patient management. As a result, the application enables the integration of the analysis data into the context of patient management. Individual findings are thus stored within the system for each patient and are therefore accessible at any time and contextually embedded in the respective examination information.

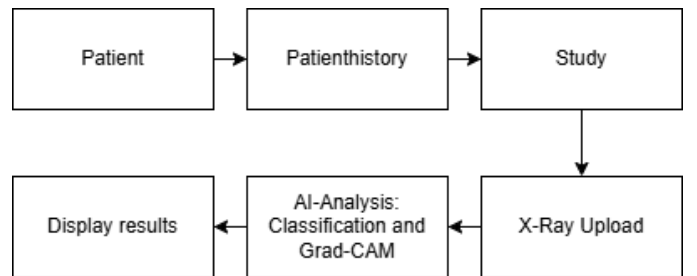


Fig. 1. The user-centered workflow for the application

The workflow consists of a total of six steps. The workflow begins with selecting a patient from the application's overview.

The patient's medical record can then be opened for the selected patient. This record contains basic information about the patient as well as the patient's medical history. In this history, examination entries¹ can be created and viewed, and these entries may contain files. When a new examination entry is created, it is possible to upload a DICOM or image file, which triggers an AI analysis. The files are thus assigned to a specific examination for a patient.

After the X-ray image has been successfully uploaded, the AI analysis begins. The application first transmits the data to another server, which performs the classification of diseases. In addition to the classification, a heatmap is also generated for each classification during this step so that it can later be displayed as an explanation of the model. After successful classification, the data is sent back to the application. The user can now view the individual analysed disease predictions and examine the visual explanation for them. The AI module thus becomes an integral part of the examination workflow and, consequently, part of patient management. The proposed workflow is illustrated in Figure 1 and consists of the following parts:

- Patient selection
- Selection or creation of an examination
- Upload of the X-ray image
- AI-assisted disease detection and generation of a visual explanation
- Display of the prediction with an explanation

IV. SYSTEM ARCHITECTURE

The system architecture was designed with the goal of minimizing risks and consistency gaps caused by architectural decisions from the outset. Another goal was to integrate image analysis and patient management into a single application. To achieve this, the machine learning services were offloaded to a separate server. The resulting system therefore consists of a web-based Spring Boot application that renders a Thymeleaf frontend. The machine learning service is hosted on a FastAPI server. The choice of Spring Boot and Thymeleaf results in server-side rendering. This is used instead of a separate JavaScript Single Page Application frontend because it reduces the complexity of the regulated software system, thereby streamlining the software architecture to be validated and minimizing security-related interface and consistency risks.

The Spring Boot application handles user interaction and the management of patient data. The ML service is responsible solely for processing the image data and delivering the results. It is accessed via a REST interface by the Spring Boot backend and can thus be used independently of the patient management system. Both components remain interchangeable with respect to one another. A diagram of the system architecture is shown in Figure 2. The architecture thus follows a strict separation between the application and the AI analysis. The user interacts with the application exclusively via the web interface.

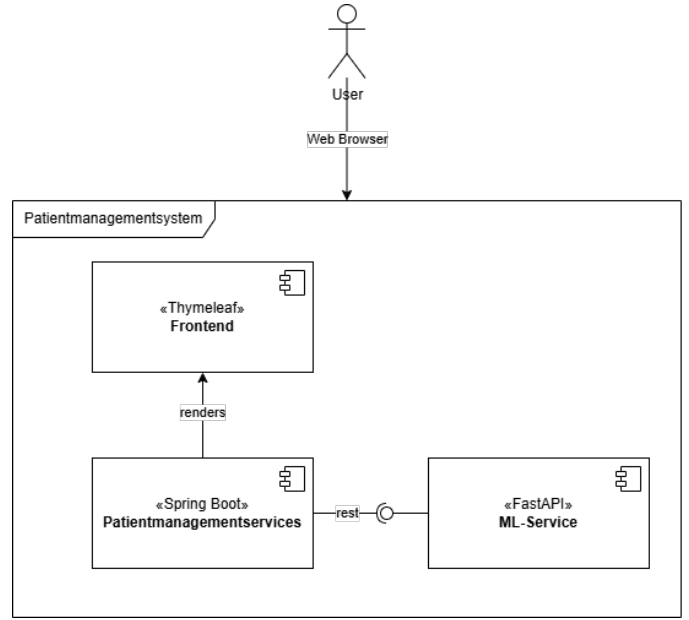


Fig. 2. The system architecture for the application

A. Patient and Study Data

The management of patient and study data, on the one hand, reflects the patient management system and, on the other hand, provides the framework within which the AI analysis is integrated. To this end, patients, examination entries, studies, and X-ray images are modeled as interdependent entities. A patient can have multiple examination entries, each of which can store files and X-ray images. The data structure thus maps the patient-centered workflow described in the concept to the data level as well.

B. Image Processing and ML Pipeline

The analysis of an X-ray image begins with uploading the image to an examination. Before storage, the incoming image format is checked and converted to a uniform PNG format. PNG and JPEG images as well as DICOM files are supported.

After successful saving, an analysis process is initiated for the image. To this end, the image is transmitted to the ML service. A detailed description of the CNN can be found in Chapter VI-A.

A defined threshold value is used to determine which findings are considered positive in the prototype. If the probability of a pathology exceeds this threshold, a Grad-CAM visualisation is additionally generated for the corresponding finding, and the result is sent back to the server. The resulting heat map is superimposed on the X-ray image and sent back to the server along with the results. The generated analysis results thus include both the classification and the associated individual probabilities, as well as the corresponding visual explanations.

¹In the application named HistoryEntry and Study

V. USER INTERFACE

The UI serves as the interface between users and the prototype. Particular emphasis is placed on the UI. The goal of the UI design was to make it as user-friendly and intuitive as possible. Therefore, it treats AI analysis and patient management not as separate entities, but as integral parts of a patient-centered workflow. The UI therefore enables both the management of patient-related data and access to examinations along with their AI analysis results. Screenshots of the application can be found in Appendix XIV.

The design is intentionally kept minimal. Important buttons are highlighted in the primary color blue (e.g., the button to open the patient record in Figure 7), while other information is organized and displayed in gray boxes (see, for example, the patient information in Figure 8) to establish a visual hierarchy. The analysis results and the Grad-CAM are also integrated. The predicted disease with the highest probability is displayed as a red dot indicating its status (see Figure 9). This allows for a quick glance at the most likely prognosis. Red is chosen here because it is a general warning signal. The Grad-CAM display, in turn, is color-coded with blue and red areas, as described in the paper. This also aligns with the overall color scheme of the UI.

When navigating between components, explicit care is taken to ensure that the paths are as simple and fast as possible, so that in everyday use, the paths are short and the application can be used quickly. The color hierarchies also ensure that navigation through the pages is guided by the colors.

VI. TECHNICAL COMPONENTS AND IMPLEMENTATION

This chapter presents the individual components of the prototype and their implementation in detail.

A. Convolutional Neural Network

The machine learning component of the prototype was developed to automatically detect abnormalities in chest X-rays. The fundamental architectural and modeling decisions were based on CheXNet by Rajpurkar et al. [39]. A DenseNet-121 model pre-trained on ImageNet was used as the foundation and further trained on the NIH Chest X-ray Dataset.

The model was implemented as a multi-label classifier for the 14 thoracic pathologies in the dataset. This allows multiple abnormalities to be predicted simultaneously for a single image. The original classification layer of the DenseNet-121 was replaced for this purpose with a classification head having 14 output values.

The official training and test sets from the NIH Chest X-ray Dataset [26] were used for training. In addition, a separate validation dataset was created from the training data. The data was split at the patient level, ensuring that no images from the same patient were included in multiple datasets.

The best model was selected based on the macro-AUC on the validation dataset. A maximum validation value of Validation-Macro-AUC = 0.8322 was achieved. Classification thresholds were then determined for each pathology based on the validation data and applied unchanged to the test data.

To improve the transparency of the model predictions, Grad-CAM [18] was also used. The visualisations generated by this method highlight the image regions relevant to the respective prediction. The model outputs were then integrated into the developed prototype and displayed there as supporting information. Grad-CAM is explained in Section VI-C.

In addition, the impact of input resolution on model performance was investigated. In addition to the 224×224 -pixel resolution used in CheXNet, a higher resolution of 448×448 pixels was considered. Both model variants were trained under otherwise identical conditions and differed only in terms of input resolution. The model with 224×224 pixels achieved a maximum validation macro-AUC of 0.7656, while the model with 448×448 pixels achieved a value of 0.8312. This corresponds to an absolute improvement of 0.0656 AUC points. Due to the improved validation performance, the input resolution of 448×448 pixels was used for the final model. The results of the comparison are shown in Table I.

TABLE I
VERGLEICH DER MODELLLEISTUNG BEI UNTERSCHIEDLICHEN
EINGABEAUFLÖSUNGEN.

Eingabeauflösung	Validation-Macro-AUC	Test-Macro-AUC	Test-Loss
224×224	0.8228	0.7956	0.5808
448×448	0.8322	0.8071	0.5511

B. Image Preprocessing

Before the CNN analyses the X-ray images, the images must first be converted into a uniform input format. Image files (PNG, JPEG) as well as the DICOM X-ray format are accepted for this purpose. DICOM and JPEG files are converted to PNG images using the dcm4che library² and then saved. This is necessary because medical image formats such as DICOM differ from conventional PNG images in terms of resolution, contrast, and image information. This ensures that the CNN only makes predictions based on images with which the network was trained. The preprocessing is designed to preserve as much of the image information relevant to diagnostic analysis as possible.

C. Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is used to visualize the machine learning model's predictions [18]. Grad-CAM makes it possible to highlight areas of the image that contributed significantly to the model's prediction. The resulting heatmap then highlights areas that had a significant influence in red and those with little influence in blue. This is intended to make the model's decisions comprehensible and assessable for medical professionals, as it allows them to see which parts of the image the CNN used to detect the disease. In the developed system, Grad-CAM thus serves as an explanatory component between the model prediction and the user and is returned by the FastAPI server.

²<https://web.dcm4che.org/dcm4che-library>

D. Cloud-Deployment

In addition to the prototypical integration of a patient management system with AI-based analysis of chest X-rays, another goal was to make the prototype available via the cloud. To make this possible, the components were subsequently deployed separately in a cloud environment. Microsoft Azure was used for the cloud deployment, as Microsoft provides one of the largest cloud infrastructures in the world [40] and therefore represents a realistic use case for such a deployment.

The Spring Boot application plays a central role within the cloud architecture. On the one hand, it provides the user interface. On the other hand, it manages all patient-related data. At the same time, it is responsible for coordinating the analysis of AI-based disease detection. The Spring Boot application's container is accessible via an external HTTP endpoint and thus serves as the sole direct access point for the user.

The machine learning service is deployed within Azure as an Azure Container App. As a result, the containers can both operate independently of one another, although the Spring Boot container has access to the machine learning container, which is visible only within Azure. This structure makes it possible, in particular, to decouple the more computationally intensive machine learning component from the actual web application. In addition to these two components, data storage in the cloud is also managed. An Azure-based file storage solution is used for this purpose, specifically for storing files. The application can then access this storage during its runtime. This ensures data consistency.

The resulting cloud architecture thus maps the logical system architecture described above to the infrastructure level. External access occurs exclusively through the Spring Boot application, while the ML service operates as an internal component of the cloud environment. Communication between the components takes place via the defined REST interface. Thanks to containerization, the developed prototype can be deployed in a reproducible manner, and the application components can be operated independently of one another.

The cloud deployment is not intended to serve as a highly scalable, production-ready cloud infrastructure for a hospital. Rather, we aim to show that a cloud deployment is possible, to demonstrate the functionality of the overall system under realistic operating conditions, and to enable the integration of the web application with AI-based image analysis as a cohesive application system.

VII. USER STUDY: INTEGRATION OF PATIENT MANAGEMENT SYSTEMS AND AI-SUPPORTED DIAGNOSIS

In addition to factors such as technical performance and compliance with regulatory requirements, usability is a key factor in the successful deployment of software systems in clinical settings. This is particularly true for systems that incorporate AI into existing clinical workflows.

A. Objective and Research Question

The objective of the user study was to evaluate the developed system from the perspective of potential medical users. The focus was on users' perceptions of the system as well as potential challenges in integrating AI-based X-ray image analysis into clinical workflows. The study is therefore intended to provide initial insights into how healthcare professionals perceive an integrated system combining a patient management system and AI-based diagnosis, and what usability challenges arise when integrating AI into clinical workflows.

B. Method

The following section explains the methodological approach.

1) *Study Design:* The study was conducted as a mixed-methods user study and included a qualitative evaluation using the think-aloud method [41] as well as a quantitative evaluation based on the User Experience Questionnaire (UEQ). All participants were asked to voice their thoughts, questions, and observations aloud while completing the tasks. This approach was intended to reveal comprehension issues, navigation difficulties, and expectations during the interaction.

The study was conducted entirely online in German, while the application was in English. Participants could use various devices, though the use of a laptop was recommended due to its superior display. Each session lasted an average of about 20 to 25 minutes. The participants' comments and observed interactions were documented in writing during the study.

All participants completed the same tasks in the same order. They were first asked to solve the tasks independently. Assistance was provided only when obvious difficulties prevented them from continuing with a task.

2) *Sample:* A total of $n = 6$ individuals with diverse medical and technical backgrounds participated in the study. A medical background was a prerequisite for participation in this study. The sample included medical assistants, medical students, and individuals with clinical experience who were recruited through personal contacts. The individual experience profiles are presented in Table II. Before the tasks began, all participants were surveyed regarding their professional experience as well as their experience with X-ray images, medical information systems (MIS), AI-powered clinical decision support systems (CDSS), and cloud-based applications.

3) *Study procedure:* The study consisted of three phases: (1) Gathering background information through an interview using custom items, (2) Completing standardized think-aloud tasks, (3) Assessing the user experience using the User Experience Questionnaire (UEQ)³.

At the beginning, background information was gathered on the participants' professional experience, as well as their prior experience with X-ray images, medical information systems, AI and decision support systems, and cloud-based applications. After providing information about their prior professional and technical experience, the participants completed

³<https://www.ueq-online.org/>

TABLE II
OVERVIEW OF STUDY PARTICIPANTS AND THEIR EXPERIENCE PROFILES

ID	Role	Gender	Age	X-ray Experience	MIS Experience	AI Experience	Cloud
P01	Medical Assistant, Dermatologist	Female	31	Low	Medium	None	No
P02	Medical Assistant, Medical Degree	Female	29	Medium	Medium	None	No
P03	Radiology Ph.D. Candidate	Male	23	Medium	Medium	Yes	No
P04	Medical student	Male	34	Moderate	Low	None	No
P05	Medical student	Female	23	Low	Moderate	Yes (scores)	No
P06	Medical assistant in geriatrics	Female	31	None	Moderate	None	No

three tasks to gain an understanding of the system’s typical workflow. They then answered a final open-ended question and filled out the User Experience Questionnaire (UEQ). The data was collected using a think-aloud protocol.

- 1) Search for the patient Anna Becker and open her medical record,
- 2) Upload a provided chest X-ray image,
- 3) Review the uploaded X-ray image based on the AI model’s analysis.

The tasks were deliberately selected to guide participants through both patient management and AI-assisted prognosis. All participants uploaded the same chest X-ray image, so they all saw the same result. While working on the tasks, all participants were asked to voice their thoughts and any difficulties they encountered. After completing the tasks, all participants were given the opportunity to provide additional comments or suggestions for improvement in an open-ended closing question.

To assess the user experience, the User Experience Questionnaire (UEQ) developed by Laugwitz et al. [42] was subsequently used. The UEQ consists of 26 items in the form of pairs of opposing terms and maps these to six dimensions of user experience. In this study, the response scale was adjusted from seven to five levels. The results of the UEQ were analysed and complement the qualitative findings from the think-aloud study.

C. Data Analysis

The user study was analysed for both the qualitative and quantitative aspects. The following sections describes the analyses and the results.

1) *Qualitative Analysis:* The qualitative data were analysed using thematic analysis based on the approach of Braun and Clarke [43]. The data consisted of the transcripts produced during the think-aloud tasks, as well as additional comments provided by the participants.

The transcripts were coded based on relevant comments, observed usability issues, misinterpretations, and suggestions for improvement. Subsequently, codes with similar content were merged and grouped into overarching themes. In the interpretation process, both the frequency of a problem and its impact on task completion and understanding of the AI outputs were taken into account.

2) *Quantitative Analysis:* For the UEQ data, a analysis was performed for each dimension using the mean (M) and variance (Var). Due to the small sample size of $n = 6$, it

is not possible to make statistical generalizations about the results.

D. Results

The following is an explanation of the results. The interview quotes presented below in English were translated from German into English, as the study was conducted in German. The translations were done by the authors themselves and are faithful to the original meaning of the statements.

1) *Qualitative Results:* The qualitative analysis identified five key thematic areas: navigation and information architecture, image upload and system feedback, AI visualisation and GradCAM, clinical context, and perceived reliability of the AI.

a) *Navigation and Information Architecture:* The participants generally quickly grasped how to search for a patient and open a patient record. Difficulties arose in particular with navigating the patient record and understanding individual terms and structural elements.

For P06, a difficulty arose when navigating within the patient record. When switching from an existing examination entry, she first looked for a way to return to the previous view. Since she could not immediately find the corresponding button, she accidentally closed the patient record and then had to reopen the patient record. This suggests that navigating between the patient overview, the patient record, and individual examination entries was not always self-explanatory.

The labels for individual elements also caused some confusion. For example, P04 questioned the meaning of the “Patient ID” as well as the options “Inpatient” and “Outpatient.” He requested additional information indicating whether a patient was receiving inpatient or outpatient care and how this information should be interpreted in a clinical context. This made it clear that not all of the terms used were unambiguously understandable without further explanation.

Furthermore, the hierarchical structure of the patient record was not immediately clear to all participants. For example, P03 did not realize until after several steps that an attachment is associated with an event. This observation shows that the relationship between investigation entries and the information or files they contain was not immediately apparent from the presentation.

Overall, the observations show that while the basic navigation was generally understandable, certain elements of the information architecture did not fully meet the participants’ expectations. Clearer labeling of navigation elements, as well

as additional explanations of technical and structural terms, could improve navigation within the patient record.

b) *Image Upload and System Feedback:* When uploading images, there were repeated difficulties in understanding the underlying process. While it was generally possible to perform the upload technically, uncertainties centered in particular on how a new X-ray image is integrated into the patient record, how existing images are handled, and how to determine whether an upload was successful.

This became particularly clear in the case of P05. After opening an existing event that already contained an uploaded X-ray image, she looked for a way to add another image. At first, she expected a process in which a new X-ray image would be requested, rather than uploading an image herself. This was evident in her statement that she wanted to request a "new [X-ray image]" (P05). In addition, it remained unclear to her whether uploading another image within an existing event would overwrite the existing image. This observation illustrates that the specific process for adding a new X-ray image did not fully correspond to her expected clinical workflow.

P03 also initially looked for the upload function in a different part of the system. He initially assumed that an X-ray image could be uploaded while creating a new event. After he couldn't find a way to do so there, he exited the process and searched for the attachments again in the overview. Only then was he able to understand the intended workflow and create an event with an associated image.

In addition to the process design, there were uncertainties regarding system feedback and supported file formats. After the upload, P01 was initially unsure whether the image had actually been saved and assumed it had been because the image was subsequently displayed. In addition, she questioned the restriction to PNG files, since X-ray images used in clinical settings may also be available in other file formats. She therefore requested that the supported file formats be more clearly identified.

Overall, the observations show that the difficulties encountered when uploading images were less related to the technical process itself and more to the expected workflow and the system's feedback. Clear confirmation of successful saving, clear labeling of supported file formats, and a more intuitive process for adding new X-ray images could reduce uncertainty during image uploads.

c) *AI visualisation and GradCAM:* The GradCAM visualisation was generally perceived as a helpful supplement to the AI diagnosis. At the same time, its significance was not immediately clear to all participants. In particular, the relationship between color highlighting, diagnostic findings, and the confidence score led to uncertainty.

This was evident, for example, in the case of P05, who, after looking at the heat map, asked: "Why are other areas red, too?". Although she generally recognized the diagnosed effusion, she could not understand why other areas of the X-ray image were also highlighted in color. She therefore considered the heat map to be too imprecise and questioned

the extent to which the visualisation made the AI's diagnostic conclusion comprehensible.

P03 also did not initially establish a direct connection between the shadowing shown and the diagnosed condition. Only after reviewing the condition again did he realize that the shadowing was also highlighted in the heat map. This suggests that the connection between the diagnostic output and the visual highlighting was not immediately apparent.

In addition, there were uncertainties regarding the meaning of the confidence value and its relationship to the color coding. P06 asked, "What does 'confidence' mean?". Later on, she assumed that the red color in the GradCAM display represented effusion and that the confidence value was related to the blue color. This interpretation did not correspond to the intended meaning of the information presented and illustrates that the various visual and numerical elements of the AI output were not clearly distinguished from one another.

In addition to interpreting the content, there were also difficulties regarding the clarity and comparability of the presentation. P01 was initially unsure which elements of the GradCAM view could be selected and needed to review the explanation to better understand the function. She also expressed a desire to display the GradCAM image at the same size as the original image so that the two images could be compared more easily. P04 initially interpreted the red zones as separate statements or findings rather than as a unified indication of a single condition. This also illustrates the difficulty of immediately grasping the meaning and classification of the color-coded markings.

In addition, there was a desire for a textual explanation of the AI's output. P02 commented: "I would like a more detailed textual explanation". The purely visual representation was thus deemed insufficient to understand the AI's decision and to compare it with one's own observations of the X-ray image.

Overall, the observations show that while the GradCAM presentation was generally perceived as a helpful supplement, its significance was not always immediately clear to the participants. In particular, the relationship between color highlighting and diagnostic findings, the significance of the confidence score, and the spatial accuracy of the heat map led to uncertainties. A clearer explanation of the visual and numerical elements, a more precise depiction of relevant image areas, and a layout that allows for better comparison between the original image and the GradCAM image could help improve the transparency of the AI output.

d) *Clinical Context:* The participants expressed a desire for AI analysis to be more closely integrated into the clinical context. In addition to the X-ray image, patient information, pre-existing conditions, medications, laboratory results, and previous examinations were considered particularly relevant for evaluating the diagnosis. In addition, further information was requested regarding other findings and possible next diagnostic steps. The AI output was therefore not viewed as isolated information, but rather as part of a broader clinical decision-making process.

e) *Perceived Reliability of AI*: The reliability of the AI was assessed based on the displayed confidence value and the agreement between the AI output and the visible findings. The displayed confidence value of 74% was sometimes perceived as low. P03 commented on this: "A 74% confidence level is quite low, considering how conspicuous it is". At the same time, there was uncertainty regarding the meaning of the confidence value. P06, for example, asked, "What does 'confidence' mean?". Later on, she also mistakenly linked the confidence value to the color coding in the GradCAM visualisation and interpreted the red and blue colors as indicating different meanings of the confidence value. This shows that not only the concept of "confidence," but also the distinction between the numerical confidence value and the visual highlighting was not immediately clear to the participant.

In contrast, agreement between the AI's output and the user's own assessment was viewed positively. Additional markings or those perceived as inaccurate led to skepticism regarding the AI's output. For example, P05 questioned why, in addition to the suspected finding, other areas of the X-ray image were highlighted in red: "Why are other areas also red?" (P05). Thus, both the substantive plausibility of the AI output and the clarity and transparency of its representation of uncertainty influenced the perceived reliability.

2) *UEQ Evaluation*: To complement the qualitative think-aloud analysis, the User Experience Questionnaire (UEQ) was used to assess the participants' subjective experience of the developed prototype. The questionnaire evaluates six dimensions of user experience: Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty.

Since the questionnaire used in this study employed a five-point response scale, whereas the standard UEQ analysis procedure is based on a seven-point scale, the collected responses were recoded prior to analysis. The original response categories were mapped to the seven-point UEQ scale as follows:

$$1 \rightarrow 1, \quad 2 \rightarrow 2, \quad 3 \rightarrow 4, \quad 4 \rightarrow 6, \quad 5 \rightarrow 7.$$

This transformation was applied consistently to all questionnaire responses. The resulting data were then entered into the UEQ Data Analysis Tool⁴. The analysis tool accounts for the polarity of the individual questionnaire items and subsequently calculates the scores for the six UEQ dimensions. All calculations are taken from this analysis tool.

Due to the small sample size, the analysis was conducted descriptively. For each dimension, the mean (M) and variance (Var) were calculated. The results should therefore be understood as exploratory observations of the participants' perceived user experience rather than as statistically generalizable estimates for a larger population. The results are displayed in table III.

The mean scores for all six UEQ dimensions were positive. The highest mean values were observed for *Attractiveness* and

TABLE III
UEQ SCALES ACROSS THE SIX DIMENSIONS.

Dimension	M	Var
Attractiveness	1.750	0.84
Perspicuity	1.542	0.99
Efficiency	1.750	1.18
Dependability	1.625	0.87
Stimulation	1.417	1.54
Novelty	1.583	2.82

Efficiency (both $M = 1.750$, with variances of $Var = 0.84$ and $Var = 1.18$, respectively), followed by *Dependability* ($M = 1.625$, $Var = 0.87$). *Novelty* reached a mean value of $M = 1.583$ ($Var = 2.82$), and *Perspicuity* achieved $M = 1.542$ ($Var = 0.99$). *Stimulation* showed the lowest mean value of $M = 1.417$ ($Var = 1.54$).

According to the interpretation provided by the UEQ analysis tool, values between approximately -0.8 and 0.8 represent a neutral evaluation, while values above 0.8 indicate a positive evaluation and values below -0.8 indicate a negative evaluation. Based on this descriptive classification, all six dimensions were evaluated positively by the participants. The results therefore indicate an overall favorable subjective perception of the prototype across the measured UEQ dimensions.

Figure 3 visualizes the scale mean scores of the six dimensions with their confidence intervals.

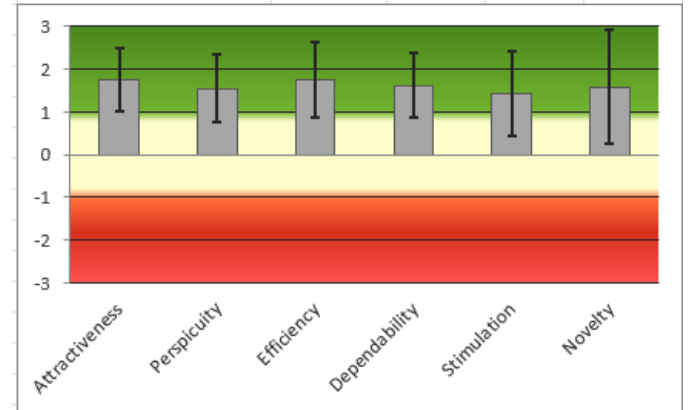


Fig. 3. Mean scores of the six UEQ dimensions with confidence intervals.

The high score for *Efficiency* suggests that the participants generally perceived the interaction with the prototype as efficient. This is consistent with the qualitative observations, where the basic patient search and opening of patient records were generally understood without major difficulties. Nevertheless, the think-aloud sessions also revealed situations in which the interaction could be improved. In particular, some participants experienced uncertainty or required additional steps during the image upload process and when navigating within the patient record. The positive UEQ score therefore reflects the overall subjective perception of efficiency and does not imply that individual interaction problems were absent.

A similar interpretation applies to *Perspicuity*, which received a positive mean score of $M = 1.542$. The qualitative

⁴fount at <https://www.ueq-online.org/>

analysis indicated that the general structure of the prototype could be understood, while also revealing specific difficulties concerning terminology and information architecture. For example, participants showed uncertainty regarding distinction between Inpatient and Outpatient. In addition, the hierarchical organization of information within the patient record was not always immediately clear. The quantitative and qualitative results therefore complement each other: while the UEQ indicates a generally positive perception of clarity, the think-aloud observations identify specific elements that may still require refinement.

The *Dependability* dimension also achieved a positive mean score of $M = 1.625$ with a comparatively low variance of $Var = 0.87$. This indicates a relatively consistent evaluation of perceived dependability among the participants. However, the qualitative results revealed uncertainty regarding the interpretation of the AI-generated output, particularly concerning confidence values and the Grad-CAM visualisation. The UEQ result should therefore be interpreted as a measure of perceived dependability of the overall interaction rather than as evidence of the objective reliability or clinical validity of the underlying AI model.

The UEQ dimensions can additionally be considered in terms of pragmatic and hedonic quality. Perspicuity, Efficiency, and Dependability represent pragmatic aspects of the interaction, whereas Stimulation and Novelty represent hedonic aspects. Attractiveness provides an overall assessment of the perceived appeal of the product. Figure 4 illustrates the UEQ results grouped according to these categories.

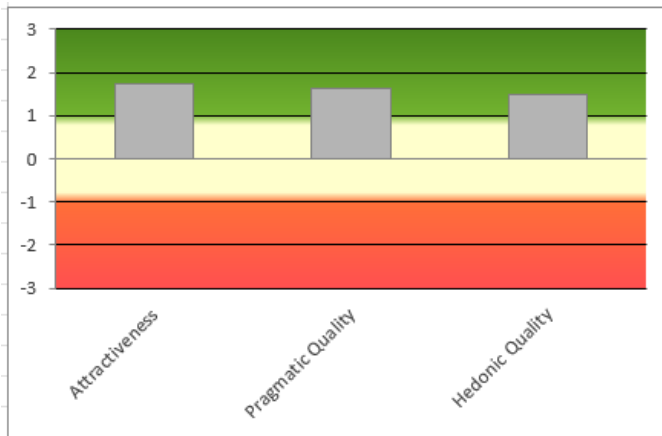


Fig. 4. UEQ scores grouped by pragmatic quality, hedonic quality, and attractiveness.

The grouped results indicate positive evaluations for both pragmatic and hedonic aspects of the prototype. The pragmatic dimensions were characterized by positive scores for Perspicuity, Efficiency, and Dependability, while the hedonic dimensions (Stimulation and Novelty) also remained positive. This suggests that the prototype was perceived not only as functional and understandable but also as providing positive experiential qualities.

The variances provide additional information about differences between individual participants. The largest variation by far was observed for *Novelty* ($M = 1.583$, $Var = 2.82$), followed by *Stimulation* ($M = 1.417$, $Var = 1.54$). Although their mean scores remained positive, the comparatively large variance indicate substantial differences in how participants perceived these hedonic aspects of the prototype. This may reflect the fact that different aspects of the system were perceived differently. While the integration of AI-supported image analysis may have appeared novel to some participants, other elements, such as patient management and conventional navigation structures, may have appeared more familiar.

Given the small sample size, the uncertainty associated with the estimated dimension means is also relevant. Figure 5 presents the UEQ dimension means together with their confidence intervals.

Confidence intervals (p=0.05) per scale					
Scale	Mean	Std. Dev.	N	Confidence	Confidence interval
Attractiveness	1,750	0,917	6	0,734	1,016 2,484
Perspicuity	1,542	0,993	6	0,794	0,747 2,336
Efficiency	1,750	1,084	6	0,867	0,883 2,617
Dependability	1,625	0,932	6	0,746	0,879 2,371
Stimulation	1,417	1,242	6	0,993	0,423 2,410
Novelty	1,583	1,678	6	1,343	0,240 2,926

Fig. 5. UEQ dimension means with confidence intervals.

The confidence intervals illustrate the limited precision of the estimated dimension means resulting from the small sample. In particular, the intervals for Novelty and Stimulation are comparatively wide, which is consistent with their high variances. The confidence intervals therefore reinforce the exploratory nature of the findings. The positive mean scores should not be interpreted as precise estimates of the user experience that would be expected in a larger population.

For additional context, the UEQ Data Analysis Tool provides a benchmark comparison with existing UEQ evaluations. Figure 6 presents the benchmark classification generated by the analysis tool.

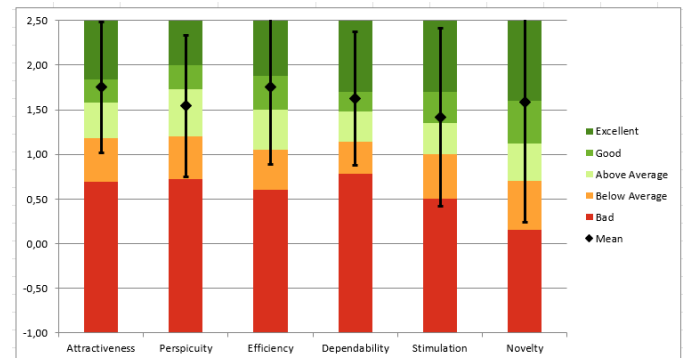


Fig. 6. UEQ benchmark comparison generated by the UEQ Data Analysis Tool.

According to the benchmark provided by the UEQ Data Analysis Tool, Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation and Novelty were classified as "Good",

whereas Perspicuity was classified as "Above Average". The benchmark data are based on 21,175 participants from 468 studies and therefore provide a substantially larger reference population than the present evaluation. However, the benchmark comparison should be interpreted cautiously in the present study because the questionnaire was originally administered using a five-point response scale and the responses were subsequently mapped to the seven-point UEQ scale. The benchmark is therefore used only as contextual information and not as evidence of statistical superiority over other systems.

a) *Integration with the Qualitative Findings:* The quantitative UEQ results complement rather than replace the qualitative findings from the think-aloud sessions. While the UEQ indicates an overall positive perception of the prototype, the qualitative analysis identifies specific interaction problems that are not necessarily reflected in the aggregated dimension scores.

For example, the positive Perspicuity score does not imply that all interface elements were immediately understandable. Instead, the think-aloud data provide a more detailed explanation of where comprehension difficulties occurred. Similarly, the positive Efficiency score indicates a favorable overall perception but does not exclude individual situations in which participants required additional steps or clarification.

This combination of findings is particularly relevant for the evaluation of the prototype. The UEQ provides a compact quantitative assessment of the subjective user experience, whereas the think-aloud analysis identifies concrete usability issues and their contextual causes. Taken together, the results suggest that the prototype was generally well received while still presenting specific opportunities for improvement, particularly in information architecture, terminology, image-upload feedback, and the presentation and interpretation of AI-supported visualisations.

Overall, the quantitative results indicate a predominantly positive perceived user experience across all six UEQ dimensions. At the same time, the relatively high variances for some dimensions, most notably Novelty and Stimulation, demonstrate that the experience was not uniform across all participants. Given the small number of valid responses and the exploratory nature of the study, these findings should be interpreted as initial evidence regarding the perceived user experience of the prototype rather than as generalizable conclusions.

3) *Additional Suggestions for Improvement:* The suggestions for improvement expressed during the think-aloud sessions and in response to the concluding question complemented the qualitative findings. In particular, participants frequently requested additional explanations of the AI output, a more precise GradCAM representation, and greater integration of clinical context information. Other suggestions concerned a comparison with the original X-ray image, as well as the display of additional findings and possible next diagnostic steps.

E. Derivation and Implementation of Improvements

Based on the results of the user study, specific areas for improvement were identified for the further development of the system. Due to the small sample size of six participants, the qualitative observations are not intended to be generalized conclusions but rather serve to highlight recurring usability issues and individual requirements.

A recurring issue concerned the clarity of the AI output. Several participants expressed a desire for additional explanations regarding the GradCAM visualisation displayed. P02, for example, stated: "I would like more text explaining this". The spatial correspondence of the color highlights was also unclear in some cases. For example P05 asked: "Why are other areas red as well?". Furthermore, there was a desire for better comparability between the original image and the GradCAM representation. P01 suggested displaying both images at the same size. These observations show that while the visual explanation provided by the AI was generally perceived as helpful, additional explanations and a more precise representation are necessary for unambiguous interpretation. The displayed confidence value also elicited varied reactions. P03 assessed the displayed 74% as a comparatively low level of confidence: "74% confidence is quite low, considering how conspicuous it is" (P03). At the same time, the meaning of the value was not immediately clear to all participants. P06 asked, "What does 'confidence' mean?" and, in some cases, mistakenly attributed the color coding of the GradCAM visualisation to the confidence value. This illustrates that the AI's numerical uncertainty and its visual explanation should be clearly distinguished from one another and explained in an understandable manner.

Another key focus was the clinical context of the AI output. Participants wanted additional information, such as details on current medications, allergies, lab results, and other clinical findings. P05 also asked about the size of a detected finding and its exact location. P03 expressed a desire for additional information on possible next diagnostic or therapeutic steps. The AI output was thus not viewed solely as an isolated image analysis, but rather in the context of other information relevant to clinical decision-making.

In addition to the AI output, there was also room for improvement in navigation and image upload. P03 initially looked for the upload function within the event creation process and only realized after several steps how to assign attachments to events. P05 initially expected a different process for adding a new X-ray image and was unsure how to handle an existing image. After uploading, P01 was initially unsure whether the image had actually been saved. These observations suggest that the upload process should be structured more clearly and that feedback regarding the successful completion of an upload should be more unambiguous.

Based on these results, selected aspects of the system were immediately revised. A key focus was on scalability and the application's user interface. The user interface was adapted to ensure that content is more easily visible and navigable

across different screen sizes and zoom levels. Not all areas for improvement identified during the study could be implemented within the scope of the project. In particular, a more extensive revision of the AI explanations, greater integration of clinical context information, and a more precise GradCAM representation represent starting points for future iterations.

The results can be summarized into five key implications for the further design of the system:

- 1) **Explainability requires additional context.** Displaying a GradCAM heatmap alone is not sufficient to make the AI output understandable and comprehensible to all users. Visual explanations should be supplemented with textual explanations and relevant clinical information.
- 2) **Trust depends on the transparency of the AI output.** Perceived reliability was influenced both by the agreement between the AI output and the visible findings, as well as by the comprehensibility of the confidence score and the spatial markings. Uncertainties and explanations provided by the model should therefore be presented transparently.
- 3) **Differences in prior experience should be taken into account.** The participants' varying levels of prior medical and technical experience led to different requirements for interpreting the AI outputs. User interfaces for clinical AI systems should therefore account for different levels of experience, for example, through understandable explanations and context-sensitive guidance.
- 4) **Linguistic and visual consistency are important.** Inconsistent terminology and representations that were sometimes difficult to interpret impaired the system's comprehensibility. Consistent language and a clear visual hierarchy can improve navigation and reduce misinterpretations. The terminology should be developed and evaluated in collaboration with individuals with medical expertise.
- 5) **Responsive design is relevant for clinical software.** Use on smaller screens sometimes led to orientation and display issues. A layout adapted to different screen sizes is therefore particularly relevant for systems used at various clinical workstations or on mobile devices.

Overall, the results are consistent with existing research on usability and clinical decision support [10], [22], [35]. At the same time, they highlight specific requirements for AI-supported clinical systems, particularly regarding the comprehensible presentation of AI explanations, the transparency of uncertainties, and the integration of AI output into the clinical context.

VIII. REGULATORY REQUIREMENTS FOR SOFTWARE AS A MEDICAL DEVICE

Even during the prototype's development, we placed particular emphasis on traceability and adherence to clean code principles, with the goal of developing a product that is as traceable as possible. The goal of the following analysis is to evaluate the resulting prototype in terms of its compliance with regulatory requirements. This analysis does not claim

to be a comprehensive conformity assessment procedure, but rather aims to identify requirements that could be relevant for the prototype's further development. Since the authors are not legal experts, no claim is made to absolute accuracy. Rather, the goal is to provide an initial analysis and evaluation of the prototype and its legal context from a programmer's perspective. To this end, the first step is to provide a legal classification of the context in which the prototype operates. The following discussion is largely based on the classification of software as a medical device within the European Union, as conducted by Hastenteufel et al. [13], as well as on the relevant requirements of the Medical Device Regulation (MDR) [14]. It should be noted, however, that regulations governing medical devices vary widely, and there are significant differences between different countries [13].

A. Medical Device Regulation 2017/745

The Medical Device Regulation 2017/745 is the central regulatory framework for software as a medical device at the European level. It regulates two of the three phases of medical device authorization: the development phase at the manufacturer and the operational phase by the provider [13], [14]. The Medical Device Regulation (MDR) thus governs the product development, authorization, production, commissioning, and operation of all medical devices within the European Union, including software, provided that such software meets the criteria for a medical device based on its intended use [13].

To determine whether software can be considered a medical device under the MDR at all, one can refer to the definition of a medical device in Article two, paragraph one of the MDR [14]. This paragraph stipulates that, for the purpose of unambiguous classification, an intended use (Articles two and 12 of the MDR) must be established for the product. This is of central importance because, on the one hand, it determines the initial decision as to whether software is a medical device and, in the subsequent (positive) process, is decisive for the classification of the software. According to Article two, paragraph 12 of the MDR, the intended use refers to the use for which a product is intended according to the manufacturer's specifications, for example in the labeling, instructions for use, or in advertising and sales information [14]. The intended use is thus not merely a technical characteristic but describes the product's overall intended use in a medical context and serves as the basis for the subsequent classification, risk assessment, clinical evaluation, and usability assessment.

If software is classified as a medical device based on its intended use, a risk-based classification is performed in accordance with Article 51 of the MDR and Annex VIII of the MDR [14]. The MDR distinguishes between four classes: I, IIa, IIb, and III. The classification thus determines the scope of requirements for the subsequent conformity assessment procedure. Regardless of the classification, there are general obligations for manufacturers that are set forth in Article 10 of the MDR. These require a risk management system, a clinical evaluation, technical documentation, a declaration of conformity, a unique device identifier (UDI), a quality management

system, a postmarket surveillance system, consideration of the country's official language, and a vigilance system. The specific form these obligations take depends in part on the product class [13], [14].

Following classification, as part of the conformity assessment procedure under Article 52 of the MDR, manufacturers must demonstrate compliance with the so-called essential safety and performance requirements (ESPRs) and thus provide evidence of the medical device's "safety and performance" as well as fulfill other general manufacturer obligations. These include, among other things, the aforementioned requirements for risk management, technical documentation, clinical evaluation, software development, and usability [13], [14]. Following a successful conformity assessment, the manufacturer may issue an EU Declaration of Conformity and affix the CE mark. Once the software has been placed on the market, additional monitoring and vigilance procedures follow. The safety and performance requirements set forth in Annex I of the MDR are specifically relevant to software. These include, among other things, the specific characteristics of software and require appropriate reliability, repeatability, and performance. Furthermore, the principles of the software life cycle, risk management, information security, and verification and validation must be taken into account during development. To implement these requirements, various standards can be consulted, including IEC 62304 for software life cycle processes, IEC 62366-1 for usability, and ISO 14971 for risk management [13]⁵.

B. Regulatory Classification

To evaluate the prototype developed here, we must first distinguish between the two functional components of the system. On the one hand, the system represents a simplified patient management system. Functions for patient management, for displaying patient history, and for managing examination and study data serve primarily to store, organize, and display patient-related information. Taken on their own, these functions therefore do not require a specific intended use, as they cannot be evaluated as software that constitutes a medical device in the strict sense [13].

On the other hand, the system offers the ability to perform an AI-supported analysis of chest X-rays. This analysis is based on patient-specific images and generates predictions that are further explained visually by a Grad-CAM. This part of the application is therefore not limited to the storage and display of data, but is intended to be specifically used for diagnostic purposes. From a regulatory perspective, the AI-based analysis of chest X-rays in particular is a component of the overall

system that may be relevant to medical devices and would need to be approved as a medical device [13]⁶.

Based on existing AI analysis capabilities, a classification in at least Risk Class IIa can be expected. However, Class IIb or III could also be possible. This depends, however, on the specific intended purpose and on how strictly Rule 11 is interpreted (Annex VIII, 6.3 MDR). Products fall into Class III if (according to Rule 11) "death or an irreversible deterioration of a person's state of health" could occur, and Class IIb if "a serious deterioration of a person's state of health" could occur. The software falls into Class IIa if it is "intended to provide information which is used to take decisions with diagnosis or therapeutic purposes" [14]. We therefore classify our prototype as at least Class IIa, although there are strong arguments for upgrading it to Class IIb or III. In our view, the two most important factors to consider here are:

- The 14 target pathologies include findings that are potentially acutely life-threatening (e.g., pneumothorax, which may require emergency drainage or surgical intervention) and findings with serious consequences in the event of misclassification (mass or nodule classified as suspected malignancy, delayed diagnosis).
- The UI design ("AI Analysis Results," traffic-light status, Confidence in %) suggests more than mere information. It leans toward "guiding clinical management," which in critical scenarios could potentially be classified as IIb.

C. Requirements for Software Development and Usability

The regulatory requirements for software as a medical device have a direct impact on the development of such software. In addition to technical functionalities, there is a particular focus on reliability, reproducibility, safety, and usability. This is not only relevant for the prototype discussed here but applies generally to any software, and specifically to any AI software, classified as a medical device.

A central aspect here is usability, as Hastenteufel et al. explain [13]. Thus, the user interface must be designed so that it can be used safely and as intended by the intended users, thereby leading to a correct interpretation of the generated results (see our usability study in Chapter VII). In the present system, this applies in particular to the presentation of AI results. The combination of classification, confidence score, and Grad-CAM visualisation was therefore designed as an integral part of the user-centered workflow.

At the same time, the demand for usability necessitates minimizing, as much as possible, potential misinterpretations and misuse of AI results by users. The user study in Chapter VII offers initial insights in this regard, providing preliminary indications for possible further improvements to the UI. For real-world medical use, however, a larger systematic study is needed to clarify how medical users interpret the displayed results and whether the presentation could lead to incorrect decisions.

⁶For a final assessment, an intended use would be necessary to determine whether and to what extent the generated results are intended to support diagnostic decisions by medical professionals

⁵There are other standards as well, but listing and explaining them here would go beyond the scope of this discussion. However, at least the following are worth mentioning in this context: DIN ISO 13485 for quality management, DIN ISO 14155 "Clinical investigation of medical devices" and DIN 62366 on the usability of medical devices

In addition, further development into a marketable medical device would require a systematic, usability-focused, and documented development process. The IEC 62366-1 standard specifies the relevant requirements for the development of medical device usability. The user study conducted as part of this paper can be understood as a first step toward identifying and evaluating usability requirements.

Furthermore, the software development carried out as part of this paper does not meet the requirements of DIN EN 62304 for a complete software life cycle process (software requirements, architecture, implementation, and verification). In particular, there are currently no systematically documented software requirements or formalized documentation of the architecture, implementation, and verification. While this is not feasible within the context of a prototype, it would be necessary for development as a medical device. This requires, among other things, classification into one of three safety classes in order to define the individual requirements of the standard. The classes are divided into Safety Class A (no injury or harm to health possible), Safety Class B (no serious injury possible), and Safety Class C (death or serious injury possible) [13]. Given the potential consequences of an incorrect disease diagnosis, particularly if this could prevent a necessary diagnosis or treatment, a classification of at least Safety Class B appears plausible for our prototype. However, a final classification would have to be determined as part of a formal risk assessment and taking into account the specific intended use of the medical device.

With regard to the challenge of adhering to the country's official language, the final prototype allows users to switch between German, Dutch, and English. For other countries, additional languages would need to be added.

D. Verification and Testing

Based on the requirements defined above, Hastenteufel et al. also describe the need to systematically validate the system's correct functionality [13]. In particular, the requirements for verification and validation make it clear that the functionality of the individual components and their interaction must be verifiably demonstrated and documented. For the developed prototype, automated unit tests were therefore written as a first step to verify the system's functionality and ensure the software's reproducibility. These tests cover key functions of data management, image processing, and communication between the backend and the ML service. It is clear, however, that significantly more extensive verification measures would be necessary for regulatory approval.

The creation of automated tests is thus not merely a measure to improve software quality, but also a preparation for the traceability and verifiability of the development process required for medical software.

E. EU AI Act and AI-Specific Requirements

In addition to the general requirements for the authorization of medical devices under the MDR, the use of artificial intelligence gives rise to further requirements and regulations.

Of particular note here is the new EU AI Act, which establishes regulations governing the development, operation, and documentation of AI systems [44].

In medical applications in particular, there is a tension to be considered here. On the one hand, there are the potential benefits of AI in supporting medical decision-making. On the other hand, the use of AI raises a wide range of issues, from safety and reproducibility to operational and ethical concerns. In this context, Anderson and Sutherland [45] emphasize the importance of trust, evaluation, and the responsible development of AI systems in healthcare, without which their long-term integration into medical processes is hardly conceivable. This is also relevant for the present prototype, as we, too, are working with a developed ML model that achieves high diagnostic accuracies in research (see Chapter II) and can be used for diagnostic support [46].

A key challenge here lies in the traceability and evaluation of AI-generated results. While traditional software always produces the same output for the same inputs, this can no longer be guaranteed for AI systems. This raises questions about the usability of AI systems in medicine, where, for example, the MDR requires reproducibility as a criterion for medical devices [13], [14]. Particularly in the case of model updates, the question arises, for example, as to the extent to which a change necessitates re-verification and re-validation (or even a new regulatory assessment).

In addition to the MDR, the EU AI Act plays another key role in the European Union. It establishes specific, additional requirements for AI systems [44], including our AI system, which is classified as a high-risk AI system (Article 6, paragraph 2, and Annex I, Section A11). High-risk AI systems impose additional requirements regarding risk management, data quality, technical documentation, logging, transparency, and human oversight. For the prototype developed here, the relationship between the MDR and the AI Act is particularly relevant, as the AI component is not considered in isolation but is used as part of an application for processing medical image data. In particular, questions of feasibility will arise in the future, as the further development of AI systems (for example, through continued training during operation) is likely to complicate requirements for reproducibility and repeatability [13]. On the other hand, it is also clear that the training and further development of such AI systems is an integral part of the development process itself, not least to further improve the safety and reliability of these systems.

IX. DISCUSSION

The results presented here demonstrate that technical, regulatory, and user-centered requirements can, in principle, be integrated within a prototype for AI-assisted X-ray analysis. It becomes clear that the development of a deployable system goes far beyond the requirements of a model. In particular, the design of the workflow and the user interface, as well as transparent explanations of model predictions, are central components of the system, without which a system with

AI integration could likely never be released. The prototype illustrates this.

The results of the user study provide initial insights into how potential users might interpret and use the prototype. The data collected show that integrating AI analysis into the workflow can make a significant contribution to the system's usability. At the same time, however, it also makes clear that high technical performance does not automatically lead to satisfied users and actual use in everyday practice, since the added value of such a system can only be realized if AI predictions are communicated clearly and comprehensibly. This is consistent with previous research, which identifies not only model performance but also, in particular, usability, workflow integration, and trust as relevant factors for clinical use.

The integration of Grad-CAM also represents an important approach to making the AI model's decisions transparent. The generated visualisations can help users understand which image regions the model took into account. However, they should not be interpreted as a standalone diagnostic rationale. Even though the user study is not representative due to its small sample size, our results suggest that more explanations may be needed for clinicians to develop lasting trust in AI predictions. This makes it clear that explainability cannot be achieved solely through the provision of technical visualisations. Rather, it requires a comprehensible presentation and interpretation of the results that aligns with the UI design patterns and presentation formats familiar to clinicians from their professional context.

The regulatory analysis and classification of our prototype also make it clear that even during the development of a prototype, requirements must be taken into account that go beyond purely technical implementations. In particular, the classification as software that is a medical device and the associated requirements for safety, documentation, and risk management can have a significant impact on the subsequent development and deployment of a system. Similarly, there are requirements, such as the need to comply with the country's official language, that must be taken into account from the very beginning of development, as introducing them later would require a significant amount of changes. The prototype thus underscores the importance of addressing regulatory requirements early on.

Overall, our results support the assumption that the successful integration of AI-based systems requires a holistic approach. Technical performance, explainability, user-friendliness, integration into existing workflows, and regulatory requirements should therefore not be considered in isolation, but rather as interconnected components of a clinically deployable system.

X. LIMITATIONS AND FUTURE WORK

This study has several limitations which must be taken into account when interpreting the results. With regard to explainability, only Grad-CAM was used. Although the method enables an intuitive visualisation of the image regions relevant

to the model's prediction, no comparison with other XAI methods was carried out. Future work could therefore compare different explanation methods and, in particular, investigate which forms of visualisation offer the greatest added value for healthcare professionals.

The user study is also limited in terms of its validity. The sample size was comparatively small and did not consist exclusively of doctors or radiologists. The results can therefore only be applied to actual clinical practice to a limited extent. In particular, an evaluation involving radiologists under realistic clinical conditions could provide further insights into the practical usability and the actual impact of the system on the diagnostic workflow. Furthermore, the study was conducted entirely online and carried out using a variety of devices. This may have led to differences in the system's display and operation. The application was in English, but the native language of all participants was German. It is therefore possible that comprehension difficulties during the study were partly attributable to a lack of English language skills. The qualitative analysis was based on notes taken during the study and not on complete transcriptions of the think-aloud sessions. As a result, individual remarks or details of interactions may have been lost. Furthermore, due to the small sample size, the quantitative UEQ results should be interpreted solely as descriptive indicators of the perceived user experience.

Furthermore, the use of the system was only examined in a single research setting. A long-term study could reveal whether perceptions of and acceptance of the system change as usage increases. In particular, a one-off study is insufficient to capture potential habituation effects, changes in trust in AI, and the actual impact on workflows. A further evaluation could also employ methods such as the NASA-TLX to examine not only subjectively perceived usability but also the mental workload involved in using the system.

Finally, the regulatory analysis does not constitute a complete legal or regulatory assessment of the developed system. Due to the complexity of the regulatory framework and the fact that specific requirements depend on the intended use and classification of the software, this study can only provide a preliminary overview. Actual authorisation or conformity assessment would require significantly more comprehensive technical, clinical and regulatory documentation.

For future research, it would therefore be beneficial to conduct a more in-depth evaluation involving a larger sample specifically comprising healthcare professionals, multiple assessment methods, and longer-term use of the system. Furthermore, the study could be carried out under real-world clinical conditions in order to determine more precisely the system's impact on workload, diagnostic decision-making and workflow.

XI. CONCLUSION

This paper examines the development of a cloud-based prototype for AI-assisted analysis of chest X-rays, taking into account regulatory requirements, user-centred design and technical specifications. A workflow is devised that combines

AI-based analysis with a patient management system and provides explanations for the model's predictions via Grad-CAM.

The results show that a holistic approach can be implemented within a single prototype. At the same time, they highlight that the clinical usability of AI systems is not determined solely by the performance of the underlying model. Rather, explainability, usability, workflow integration and regulatory requirements must be taken into account during the development phase if such systems are to find a permanent place not only in academia but also in real-world clinical practice.

XII. OPEN SCIENCE

We would like to encourage readers to reproduce and build on our results. Our code for the web-based application and the machine-learning module as well as the results of the user study are available as open-source software at <https://github.com/AnnaLiviaMartin/project-medical-device>.

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XIV. APPENDIX I: SCREENSHOTS

The screenshot displays the 'Patient Overview' page in the MedIC system. The page is divided into a left sidebar and a main content area. The sidebar includes the MedIC logo, language selection (German, English, Spanish), and a 'Patient Management' button. The main content area features a search bar, a dropdown menu for filters, and a 'Search' button. Below the search bar is a table listing patients. The table has three columns: Patient, Address, and Emergency Contact. The first patient listed is Anna Becker, 38 years old, female, inpatient, living at Ringstrasse 12, 65183 Wiesbaden, with emergency contact Peter Becker (Spouse, +49 611 1234567). The second patient is Michael Tan, 50 years old, male, inpatient, living at Bahnhofstrasse 5, 60329 Frankfurt am Main, with emergency contact Lena Tan (Daughter, +49 69 9876543). On the right side of the main content area, there is a 'Selected Record' for Anna Becker, showing her status as Inpatient, last visit on 23 Aug 2026, next appointment on 30 Aug 2026, ward/unit as Pulmonology - Ward 3, address as Ringstrasse 12, 65183 Wiesbaden, and emergency contact as Peter Becker (Spouse, +49 611 1234567). An 'Open Record' button is located at the bottom of the selected record section. A 'New Patient' button is also visible in the top right corner of the main content area.

MedIC

LANGUAGE

German English Spanish

Patient Management

PATIENT MANAGEMENT

Patient Overview

Overview of master data, address information, and emergency contacts.

New Patient

Search by name, ID, city, or emergency contact

All

Search

PATIENT	ADDRESS	EMERGENCY CONTACT
Anna Becker #1 Inpatient 38 years · Female	Ringstrasse 12 65183 Wiesbaden	Peter Becker Spouse · +49 611 1234567
Michael Tan #2 Inpatient 50 years · Male	Bahnhofstrasse 5 60329 Frankfurt am Main	Lena Tan Daughter · +49 69 9876543

SELECTED RECORD Inpatient

Anna Becker

#1 · attending physician: Dr. Sarah Klein

STATUS Inpatient	LAST VISIT 23 Aug 2026
NEXT APPOINTMENT 30 Aug 2026	WARD / UNIT Pulmonology · Ward 3
ADDRESS Ringstrasse 12, 65183 Wiesbaden	EMERGENCY CONTACT Peter Becker · Spouse · +49 611 1234567

Open Record

Imprint

Privacy Policy

Fig. 7. Screenshot of the patient overview page.

Becker · MediC

← → ↺

http://localhost:3080/patients/1

VPN

MediC

LANGUAGE



Patient Management

← Back to Patient Overview

EditDelete

PATIENT RECORD

Anna Becker

Complete patient information, treatment history, medications, and scan access.

Inpatient

PATIENT DETAILS

Patient ID
#1

Age
38years

Gender
Female

Diagnosis
Community-acquired pneumonia

Doctor
Dr. Sarah Klein

Ward / Unit
Pulmonology - Ward 3

Last Visit
23 Aug 2026

Next Appointment
30 Aug 2026

ADDRESS & CONTACT

Address
Ringstrasse 12, 65183 Wiesbaden

Emergency Contact
Peter Becker · Spouse · +49 611 1234567

Insurance
Techniker Krankenkasse · TK-88213-A

CLINICAL HISTORY

Treatment Timeline

1 entries + Add Event

23 Aug 2026

Admission - suspected pneumonia

Patient presented with fever, cough and shortness of breath. Chest X-ray ordered.

DOCTOR
Dr. Sarah Klein

DEPARTMENT
Pulmonology

Visit

LINKED SCANS · 1 IMAGE(S)

Chest X-ray · xray_images/698c44be-ba9c-4655-8046-bd1ec498ad32.png.png

+ X-ray

Open Scan

CURRENT MEDICATION

Amoxicillin
500mg · 3x daily

Paracetamol
500mg · as needed

Imprint

Privacy Policy

Fig. 8. Screenshot of the patient details page.

Admission - suspected pneumonia

http://localhost:8080/patients/1/history/1/scan/1

VPN

MediC

LANGUAGE



Patient Management

← Back to Patient Record

23 Aug 2026

Visit

Admission - suspected pneumonia

Patient presented with fever, cough and shortness of breath. Chest X-ray ordered.

DOCTOR
Dr. Sarah Klein

DEPARTMENT
Pulmonology

ATTACHMENTS

+ X-ray

+ Document/Image

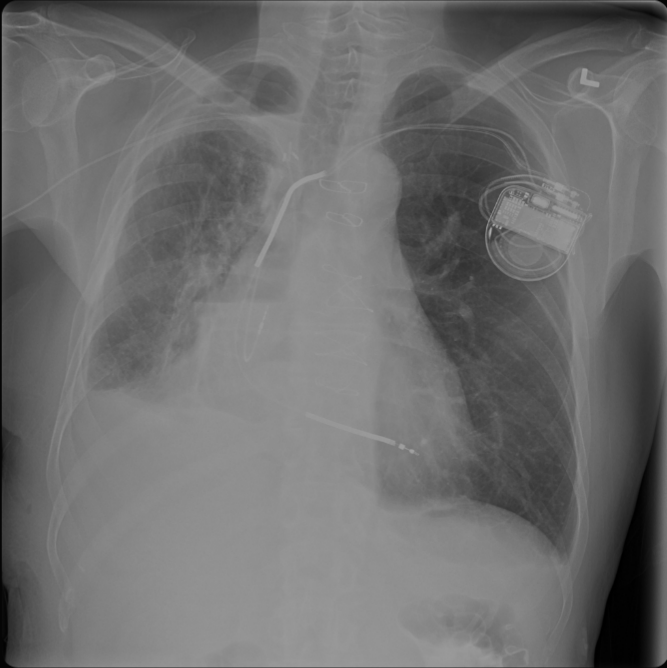
00000013_015.png

399 KB

X

Original X-ray

GradCAM - AI-Decision-Explanation



Uploaded 13 Sep 2026 12:11

Showing the original X-ray image.

AI Analysis Results

Anna Becker

Status

CONFIDENCE

74.2 %

• Effusion

SUGGESTED CLINICAL FINDINGS

Each finding below has its own GradCAM heatmap. Select a finding, then use the toggle above the image to switch between the original X-ray and its GradCAM overlay.

Effusion

74.2%

Pleural_Thickening

69.9%

Imprint

Privacy Policy

Fig. 9. Screenshot of the AI analysis page.